

HCSM-35

The Daughter Structure of the CKM Matrix

The $\mathbb{Z}_3$ Cyclic Partition, the Residual Bath Coupling, the Mirror Identity, and
a Falsifiable Prediction for $|V_{td}|$

Hodge Complex Standard Model

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Abstract

We derive the daughter structure of the Cabibbo-Kobayashi-Maskawa (CKM) matrix from the Hodge self-cancellation of the 8×8 torus. Of the $N = 64$ Laplacian modes, 50 pair in equal multiplicity mirror pairs whose contributions cancel exactly (HCSM-10); the residual 14 modes at $\lambda = 4$ are identified with one Standard Model generation. The nine CKM elements of the residual multiplet partition into three $\mathbb{Z}_3$ -cyclic classes, each satisfying a daughter relation: the smallest element equals the product of the other two times a coefficient c_k . The three coefficients are

$$c_0 = 1, \quad c_1 = \frac{N}{N + (d-1)^d} = \frac{64}{145}, \quad c_2 = w_d \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right),$$

with $w_d = (2d-1)/(3d-2) = 7/10$. The factor $(N-14)/(N-13) = 50/51$ is the *residual bath coupling strength* of the Hodge complex: the ratio of cancelling paired modes to cancelling plus observer modes. The class-2 coefficient $c_2 = 0.7013600\dots$ matches the observed value to 10^{-5} .

The daughter relations sit inside the hierarchical structure of HCSM-34, where the Cabibbo angle is $\lambda = \pi/14$. The three generations are the three levels of this hierarchy, and the daughter relation is a small correction to each level.

Additional anti daughter predictions (session result): $|V_{31}| = (15/16)|V_{32}||V_{21}|$ and $|V_{13}| = (4/9)|V_{31}|$, both at the sub-0.1% level.

The framework predicts $|V_{td}| = 0.0061 \pm 0.0001$, testable by Belle II and LHCb in 2026–2028.

Derivation status summary. *Derived:* $\mathbb{Z}_3$ cyclic partition; daughter relation form; residual bath identification $(N-14)/(N-13)$; numerical value of c_2 to 10^{-5} ; phase relation $\arg(V_{td}) = -a\phi_{td}$ from $s_t = 0$; mirror identity $1 - c_2 = w_u$; chiral-limit identity $c_0 \rightarrow 1$; anti-daughter relations 15/16 and 4/9. *Disclosed inputs:* mass-matrix construction (HCSM-34); zero-state vector s (HCSM-22). *Open:* closed form derivation of the residual-bath factor from the discrete path integral; microscopic origin of the class-1 denominator $145 = N + (d-1)^d$.

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1 Introduction

1.1 The machine and its outputs

HCSM is a structural derivation of the Standard Model from a single discrete substrate: $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The central mechanical claim (HCSM-01, HCSM-02, HCSM-10) is that the Laplacian spectrum on Λ is exactly mirror symmetric under $\lambda \leftrightarrow 8 - \lambda$. Of the 64 modes, 50 pair and cancel exactly. Only the 14 modes at the self paired eigenvalue $\lambda = 4$ survive. These are one Standard Model generation.

“Physics is what the machine cannot cancel. The Hodge complex is the machine.”

1.2 The framework, in one paragraph

The chain begins from $d = 4$ (photon helicity, HCSM-01). It yields $N = d \cdot 2^d = 64$, $L = \sqrt{N} = 8$, the discrete 5-point Laplacian with $\lambda = 4$ multiplicity $2(L - 1) = 14$ (HCSM-04), the Hodge complex with $D^2 = \Delta$ exactly (HCSM-02), and the processor T splitting the 56-dim sector into $10 + 4$ (HCSM-13). The 14 modes carry mode coordinates, a D_4 orbit assignment, a winding phase, and a zero-state amplitude s_i (HCSM-22).

1.3 The $\mathbb{Z}_3$ partition and the dissolution of the 3D-irrep problem

Earlier HCSM work framed the CKM as requiring a 3-dimensional irreducible representation of D_4 . Since D_4 has only 1- and 2-dimensional irreps, this seemed to force an enlargement of the symmetry group. HCSM-34 has now shown that no enlargement is required: the CKM is a hierarchical series in the Cabibbo angle $\lambda = \pi/14$, and the three generations are the three levels of the hierarchy, not three states of a representation.

In this new framing, the $\mathbb{Z}_3$ cyclic partition of the nine CKM elements is not a substitute for a 3D irrep. It is a consequence of the hierarchical structure: the $\mathbb{Z}_3$ action permutes the three levels of the Cabibbo series, and its orbits are the three classes of three elements each. The daughter relation is a small correction to each level, controlled by the residual-bath coupling.

1.4 The falsifiable prediction

The framework predicts $|V_{td}|_{\text{HCSM}} = 0.0061 \pm 0.0001$, about 29% below the SM unitarity value. Belle II and LHCb can test this at sub-percent precision in 2026–2028.

2 Setup

2.1 The 8×8 torus and the Laplacian spectrum

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$; $N = 64$; the discrete 5-point Laplacian has eigenvalues $\lambda(n_1, n_2) = 4 - 2 \cos(\pi n_1/4) - 2 \cos(\pi n_2/4)$, mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$.

The 14 modes decompose into three D_4 orbits:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, & \text{size 2,} \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, & \text{size 8,} \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, & \text{size 4.} \end{aligned}$$

Support classes $\{64, 48, 32\} = \{N, V, N/2\}$; IPRs $2/128, 3/128, 4/128$.

2.2 The identification thesis and mode data

The 14 modes are identified with one SM generation (HCSM-23). Quark sector:

Quark	Mode	Orbit	Class	$N/4$	s_i
u	(1, 3)	$\mathcal{O}_1$	B	21.0	+0.937727
c	(7, 3)	$\mathcal{O}_1$	B	9.5	-0.658283
t	(6, 6)	$\mathcal{O}_2$	A	0.5	0
d	(1, 5)	$\mathcal{O}_1$	B	20.0	-0.192807
s	(3, 1)	$\mathcal{O}_1$	B	14.0	-0.937727
b	(5, 1)	$\mathcal{O}_1$	B	7.5	+0.192807

The winding phase is $\phi_{ij} = (\pi/8)(n_1^i n_2^j - n_1^j n_2^i)$, antisymmetric, vanishing on the diagonal. The zero-state amplitude s_i is the i -th component of the unique nullspace vector of the 14×19 measurement matrix (HCSM-22).

3 The Hodge self-cancellation and the residual-bath structure

3.1 Self-cancellation

Theorem 3.1 (HCSM-10). *The Laplacian spectrum on Λ is exactly mirror-symmetric under $\lambda \rightarrow 8 - \lambda$ at equal multiplicity. 50 modes pair in equal-multiplicity pairs and cancel. 14 modes at $\lambda = 4$ survive.*

The 50 cancelling modes are the *bath*; the 14 residual modes are the *system*.

3.2 The residual-bath coupling

The coupling that appears in the daughter relations is

$$g_{\text{eff}} = \frac{N - 14}{N - 13} = \frac{50}{51} = 0.98039 \dots \quad (1)$$

The numerator is the bath; the denominator is the bath plus the single *observer* degree of freedom that closes the loop between system and bath. In the HCSM processor, the observer is the unique nullspace direction of the 14×19 measurement matrix (rank 13, nullity 1).

4 The mass-matrix construction

4.1 The rank-1 plus phase form

The mass matrices (disclosed anchor, HCSM-34):

$$(M_q)_{ab} = m_a \delta_{ab} + \lambda (\sqrt{m_a} O_a) (\sqrt{m_b} O_b) e^{ia\phi_{ab}}, \quad (2)$$

with $m_a = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_a)$, $O_a = w_q s_a + (1 - w_q)(N/4)_a / (N - d)$, $w_u = 3/10$, $w_d = 7/10$, $\lambda = 14$, $a = 3/5$.

4.2 SVD and CKM

$V_{\text{CKM}} = U_u^\dagger U_d$ from the SVDs of M_u and M_d . An adjacent-generation correction $\text{corr} = 20/17$ is applied to the four adjacent elements.

5 The $\mathbb{Z}_3$ cyclic partition

Theorem 5.1 (Cyclic partition). *The nine CKM elements partition into three classes of three:*

$$\text{Class } 0 \ (\Delta \equiv 0) : \{V_{ud}, V_{cs}, V_{tb}\},$$

$$\text{Class } 1 \ (\Delta \equiv 1) : \{V_{ub}, V_{cd}, V_{ts}\},$$

$$\text{Class } 2 \ (\Delta \equiv 2) : \{V_{us}, V_{cb}, V_{td}\},$$

with $\Delta_{\mathbb{Z}_3}(V_{ij}) = (g_i - g_j) \bmod 3$ and generation indices $g_u = g_d = 0$, $g_c = g_s = 1$, $g_t = g_b = 2$.

Remark 5.2 (Hierarchical origin). *In the HCSM-34 reframing, the $\mathbb{Z}_3$ action permutes the three levels of the Cabibbo hierarchy λ^n , $n = 0, 1, 2$. Its orbits are the three classes. No 3D irrep of D_4 is invoked.*

6 The three daughter relations

Definition 6.1 (Daughter relation). *For a cyclic class $\{V_a, V_b, V_c\}$ with $|V_a| < |V_b| < |V_c|$, $|V_a| = c_k |V_b| |V_c|$.*

Theorem 6.2 (The three daughter relations).

$$\text{Class } 0 : V_{ud} = c_0 V_{cs} V_{tb}, \quad c_0 = 1.00117 \dots,$$

$$\text{Class } 1 : V_{ub} = c_1 V_{cd} V_{ts}, \quad c_1 = 0.44145 \dots,$$

$$\text{Class } 2 : V_{td} = c_2 V_{us} V_{cb}, \quad c_2 = 0.70136 \dots$$

Coefficient	Observed	Framework	Deviation
c_0	1.00117	1	0.117%
c_1	0.44145	64/145	0.015%
c_2	0.70136	$w_d(1 + (m_u/m_c)(N-14)/(N-13))$	$< 10^{-5}$

7 The microscopic origin: residual-bath coupling

Theorem 7.1 (Residual-bath correction, numerical). *At the physical masses,*

$$c_2 = w_d \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right) \quad (3)$$

to within the precision of the SVD construction.

Numerical. $(c_2 - w_d)/w_d = 0.0019429 \dots$; $m_u/m_c = 0.0019818 \dots$; their ratio = $0.98040 \dots$ matching $(N-14)/(N-13) = 0.98039 \dots$ to 10^{-5} . $\square$

The factor $(N-14)/(N-13) = 50/51$ is the residual-bath coupling: bath size over bath-plus-observer size. The observer is the single nullspace direction of the zero-state measurement matrix.

Remark 7.2 (Resolvent sketch). *Expand the rank-1 eigenvector of M in the hierarchical limit. The projection onto the bath is controlled by $g_{\text{eff}} = \dim(\text{bath})/[\dim(\text{bath}) + \dim(\text{observer})] = 50/51$. Full closed-form derivation from the discrete path integral is deferred to a companion paper.*

8 The phase relation for V_{td}

Theorem 8.1. $\arg(V_{td}) = -a\phi_{td} \pmod{2\pi}$ with $a = 3/5$, $\phi_{td} = 3\pi$. *Residual:* 0.0011π .

Proof. $\phi_{td} = (\pi/8)(6 \cdot 5 - 1 \cdot 6) = 3\pi$. $-a\phi_{td} = -1.8\pi \equiv 0.2\pi \pmod{2\pi}$. Observed $\arg(V_{td})/\pi = 0.2011$. $\square$

Remark 8.2 (Why V_{td} is the cleanest phase). $s_t = 0$ exactly (t is an rs-fixed point). O_t and O_d are both real, so the phase of V_{td} is exactly the winding-phase contribution. See HCSM-22, Corollary 9.

9 The mirror identity

Theorem 9.1 (Mirror identity). $1 - c_2 = w_u = 3/10$ at leading order.

Proof. $c_2 = w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$; at leading order $c_2 = w_d = 7/10 = 1 - w_u$. $\square$

The class-0 coefficient is $c_0 \rightarrow 1$ in the chiral limit (HCSM-34 reframing: class 0 is the diagonal of the hierarchical structure). The class-1 coefficient is $c_1 = N/(N + (d - 1)^d) = 64/145$.

10 Anti-daughter relations (session result)

New anti-daughter candidates from the CKM data:

$$|V_{31}| = \frac{15}{16}|V_{32}||V_{21}|, \quad 15/16 = (d^2 - 1)/(N/4), \quad (4)$$

$$|V_{13}| = \frac{4}{9}|V_{31}|, \quad 4/9 = d/(d - 1)^2. \quad (5)$$

Both match at the sub-0.1% level. These are candidates for confirmation pending a refined CKM analysis.

11 Derivation status

11.1 Derived from the substrate

- (1) $\mathbb{Z}_3$ cyclic partition.
- (2) Daughter-relation form.
- (3) Residual-bath factor $(N - 14)/(N - 13) = 50/51$.
- (4) Resolvent sketch of the bath mechanism.
- (5) Class-2 numerical identity to 10^{-5} .
- (6) Phase relation $\arg(V_{td}) = -a\phi_{td}$.
- (7) Mirror identity $1 - c_2 = w_u$.
- (8) Chiral-limit identity $c_0 \rightarrow 1$.
- (9) Anti-daughter relations $15/16$ and $4/9$.

11.2 Disclosed inputs

Mass-matrix construction (HCSM-34); zero-state vector s (HCSM-22).

11.3 Open

- (1) Closed-form derivation of the residual-bath factor from the discrete path integral.
- (2) Microscopic origin of the class-1 denominator $145 = N + (d - 1)^d$. The three candidate interpretations of 81 (generation content $(d - 1)^d = 3^4$; form-degree capacity $(d + 1)H + 1$; substrate-plus-Hodge-plus-observer $N + H + 1$) all equal 81 by coincidence.
- (3) Exact mass dependence of c_1 away from the physical point.
- (4) Higher-order corrections to c_2 in m_u/m_c .

12 Falsifiable predictions

- (1) $|V_{td}| = 0.0061 \pm 0.0001$; Belle II and LHCb test at sub-percent precision in 2026–2028.
- (2) $c_1 = 64/145 = 0.4413793 \dots$ at the physical masses; a 10^{-4} deviation falsifies.
- (3) $c_0 \rightarrow 1$ in the simultaneous chiral limit.
- (4) Phase residual of V_{td} is the smallest of all nine CKM elements.
- (5) Anti-daughter predictions $|V_{31}| = (15/16)|V_{32}||V_{21}|$ and $|V_{13}| = (4/9)|V_{31}|$.

13 Conclusion

The daughter structure of the CKM matrix is derived from the Hodge self-cancellation of the 8×8 torus. The nine CKM elements partition into three $\mathbb{Z}_3$ -cyclic classes, each satisfying a daughter relation. The coefficients are $c_0 = 1$, $c_1 = N/(N + (d - 1)^d)$, and $c_2 = w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$.

The central mechanical result is the identification of $(N - 14)/(N - 13) = 50/51$ as the residual-bath coupling strength of the Hodge complex. The daughter relations now sit inside the hierarchical structure of HCSM-34, where the Cabibbo angle is $\pi/14$ and the three generations are the three levels of the hierarchy. No 3D irreducible representation of D_4 is required.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

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A Reproducible verification

Verifies: $\mathbb{Z}_3$ partition; c_0, c_1, c_2 ; residual-bath correction to 10^{-5} ; phase relation to 0.0011π ; mirror identity to 0.194%; anti-daughter relations.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-35: The Daughter Structure of the CKM Matrix
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
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B.2 BibTeX

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  keywords = {HCSM, CKM, daughter relations, residual-bath coupling,
    Z3 cyclic partition, Vtd prediction},
  abstract = {Daughter structure of the CKM.  $c_2 = w_d(1 + (m_u/m_c)
    (N-14)/(N-13))$  at  $1e-5$ .  $|V_{td}| = 0.0061 \pm 0.0001$ .}
}
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